

MB 551 : Causal Analytics for Business Decision Making

Course Code : MB 551

Course Name : Causal Analytics for Business Decision Making

L-T-P-C : 2-0-0-2

Intended for : MBA

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Foundations of Causal Analytics:** Why and what of Causal effects and causal inference, Describing variables and relationships, Finding and using Causes, Correlation vs Causation, Causation without Correlation, Probabilities and Causation, Evidence and Causes. (4 Hours)
- **Causal Diagrams- Drawing Your Assumptions:** Assumptions about data for Causal Inference, Levels of Causation, Interventions and Counterfactuals, Data to Graphs, Drawing Causal diagrams, Moderators in Causal Diagrams, Causal Paths and their use to test the diagram, Confounding and Deconfounding, Measuring Causality, Granger Causality, and Causality Checklist. (4 Hours)
- **Business Experiments- Data Designing:** Experimental and Non-Experimental Data, Characteristics of Business Data, Data Generating Process, The magic of randomization, Random vs, Non-random data, Randomized Controlled Trials, Randomized Treatment Assignment and Causal Inference, Randomized vs. Observational Studies, Planned Experiments and Quasi-Experiments, Treatment Effects Estimation on Business Data. (6 Hours)
- **Tools for Experimentation:** Matching, Difference-in-Differences, Regression, , Simulation, Phased Rollouts, and Sensitivity Analysis in Observational Study. (6 Hours)
- **Opportunistic Data:** Defining and navigating Opportunistic Data, Anticipating and Influencing Business Outcomes, Causality with Opportunistic Data. (4 Hours)
- **Natural Experiments-Discontinuities and Instrument Variables:** Business and Natural Experiments, Analysing Natural Experiments, Difference-in-Differences in Natural Experiments, Regression Discontinuity. (4 Hours)
- **Explanation and Action: What Caused What:** Finding Causes, Explanation with Uncertainty, Automating Explanation, Evaluating Causal Claims, to Decisions. (2 Hours)

Textbooks:

1. NA

References:

1. Judea Pearl and Dana Mackenzie, **The Book of Why: The New Science of Cause and Effect**, Penguin Books, 2018.

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2. Paul R. Rosenbaum, **Causal Inference**, MIT press, 2023.
 3. Samantha Kleinberg, **Why-A guide to finding and using Causes**, Shroff Publishers and Distributors Pvt. Ltd., 2019
 4. Jefferey, T. Prince and Amarnath Bose, **Predictive Analytics for Business Strategy**, McGraw Hill Education (India) Private Ltd., 2021
 5. Jonas Peters, Dominik Janzing, and Bernhard Scholkopf, **Elements of Causal Inference: Foundations and Learning Algorithms**, The MIT Press, 2017.